

Two key ingredients found in many soaps are PEG 150 distearate and ethylhexyl glycerin. Both can be synthesized from glyceryl tristearate (Figure 1).

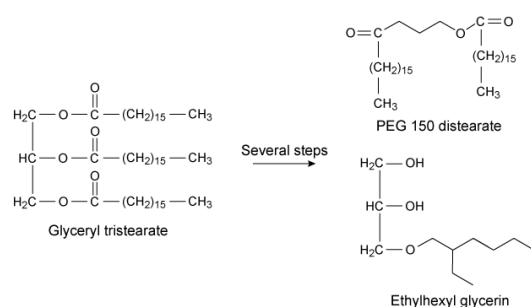


Figure 1 Formation of PEG 150 distearate and ethylhexyl glycerin from glyceryl tristearate

The first step in this process requires the saponification of glyceryl tristearate. The resulting stearate molecules can then react with PEG 150 in an acid-catalyzed Fischer esterification to form PEG 150 distearate. The glycerol produced during saponification is used as a nucleophile to form ethylhexyl glycerin through a base-catalyzed S_N2 reaction with an ethylhexyl halide.

Studies have shown that hand washing with soap can substantially reduce the number of bacteria present on hands and that washing hands before preparing food is particularly effective in reducing illness. Researchers examined the effect of washing for 20 seconds with different amounts of soap containing PEG 150 distearate and ethylhexyl glycerin and found an inverse correlation between the volume of soap used and the amount of bacteria recovered from the hands. The data are shown in Figure 2.

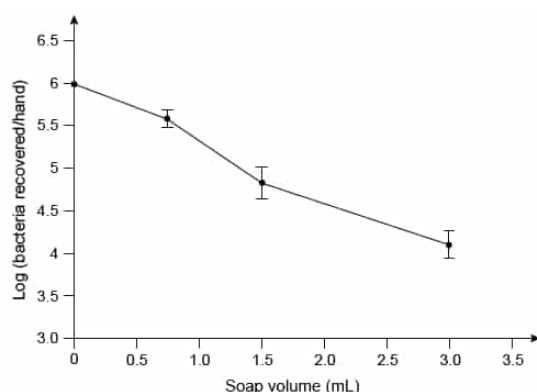
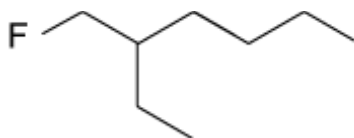


Figure 2 Bacteria recovered from hands after washing with varying amounts of soap

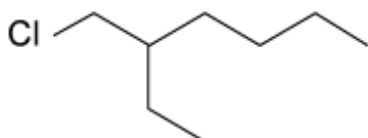
Ethylhexyl glycerin could be most efficiently synthesized from glycerol and which of the following?

☐ A.



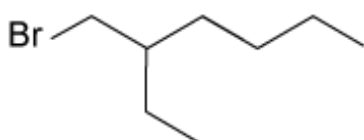
[25%]

☐ B.



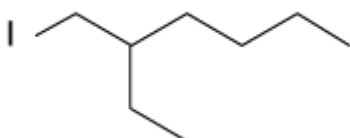
[9%]

☐ C.



[9%]

✓ ☒ D.

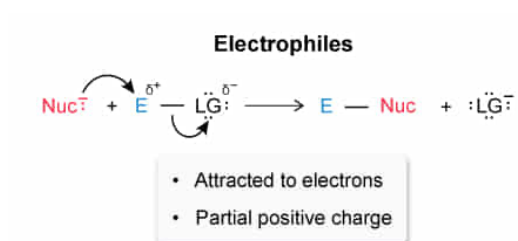


[55%]

Correct

56% answered correctly

Explanation:



© UWORLD

Bimolecular nucleophilic substitution (S_N2) reactions occur when a **nucleophile** donates electrons to an electrophile with a **good leaving group**. A leaving group is removed from a molecule when the electrons in the bond are transferred to an atom in the leaving group.

Therefore, a good leaving group must be able to **stably carry the negative charge** conferred by these electrons.

The **halogens** are generally good leaving groups because they can form stable anions. **Larger halogens** are **better leaving groups** than small halogens because they can spread the

negative charge over a larger surface area. Once the reaction has occurred, the nucleophilic atom is attached to the electrophilic molecule in the position where the leaving group was, and the leaving group is displaced.

The alcohol groups in glycerol are reasonably nucleophilic and are made more nucleophilic when deprotonated under basic conditions. They can form 2-ethylhexyl glycerin by attacking any 2-ethylhexyl group with a good leaving group at carbon 1. Of the options shown, iodide is the best leaving group because it is the largest halide and can best stabilize the negative charge upon anion formation. Therefore, 2-ethylhexyl glycerin would be most efficiently synthesized from glycerol and 2-ethylhexyl iodide (**Choice D**).

(Choice A) Fluoride is a small atom and a relatively poor leaving group. Therefore, 2-ethylhexyl fluoride would be less efficient than the other 2-ethylhexyl halides in the synthesis of 2-ethylhexyl glycerin.

(Choices B and C) These structures both contain good leaving groups, but bromide and chloride are smaller than iodide and have smaller surface areas to delocalize the negative charge when displaced. Therefore, 2-ethylhexyl chloride and 2-ethylhexyl bromide would not form 2-ethylhexyl glycerin as efficiently as 2-ethylhexyl iodide would.

Educational objective:

Bimolecular nucleophilic substitution (S_N2) reactions occur when a nucleophile donates electrons to an electrophile and the electrophile loses a leaving group. Good leaving groups must be able to stably carry the electrons they receive. Halogens are generally good leaving groups, and large halogens are better leaving groups than small halogens.

Time spent: 0 sec

Subject:
Organic Chemistry

QID: 403689

System:
Introduction to Organic Chemistry

Two key ingredients found in many soaps are PEG 150 distearate and ethylhexyl glycerin. Both can be synthesized from glyceryl tristearate (Figure 1).

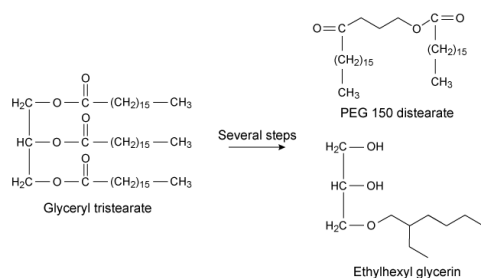


Figure 1 Formation of PEG 150 distearate and ethylhexyl glycerin from glyceryl tristearate

The first step in this process requires the saponification of glyceryl tristearate. The resulting stearate molecules can then react with PEG 150 in an acid-catalyzed Fischer esterification to form PEG 150 distearate. The glycerol produced during saponification is used as a nucleophile to form ethylhexyl glycerin through a base-catalyzed S_N2 reaction with an ethylhexyl halide.

Studies have shown that hand washing with soap can substantially reduce the number of bacteria present on hands and that washing hands before preparing food is particularly effective in reducing illness. Researchers examined the effect of washing for 20 seconds with different amounts of soap containing PEG 150 distearate and ethylhexyl glycerin and found an inverse correlation between the volume of soap used and the amount of bacteria recovered from the hands. The data are shown in Figure 2.

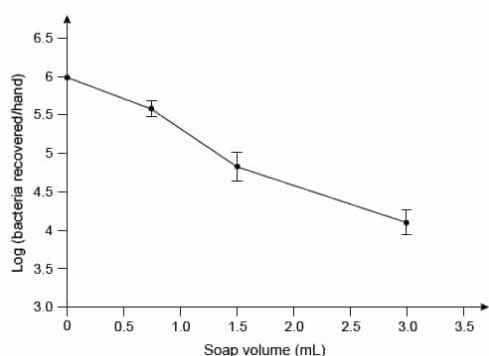
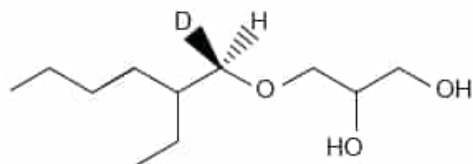


Figure 2 Bacteria recovered from hands after washing with varying amounts of soap

Researchers produced ethylhexyl glycerin using an ethylhexyl halide that contained a single deuterium atom in place of one hydrogen atom. The reaction between the ethylhexyl halide and glycerol generated the product shown below.



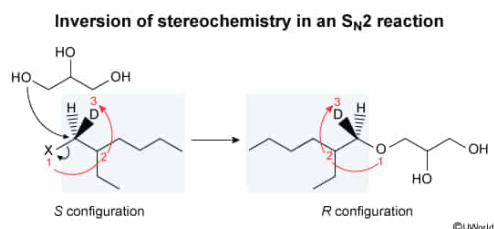
Prior to the reaction, what was the configuration of the carbon bonded to deuterium?

- ✓ ☒ A. *S*, because the configuration is inverted by the reaction [65%]
☐ B. *R*, because the configuration is inverted by the reaction [16%]
☐ C. *S*, because the configuration is preserved by the reaction [4%]
☐ D. *R*, because the configuration is preserved by the reaction [13%]

Correct

65% answered correctly

Explanation:



An S_N2 reaction is a single-step reaction in which a nucleophile reacts with an electrophile, and a leaving group is simultaneously released from the electrophile. If the electrophile in an S_N2 reaction is a **chiral center**, its stereochemistry will always be **inverted** by the reaction: an electrophile in the *S* configuration prior to the reaction will adopt an *R* configuration after the reaction, and vice versa if the nucleophile has the same priority ranking as the leaving group.

To determine whether a chiral center is *R* or *S*, **prioritize each substituent** in order of atomic number. If two substituents have the same atomic number but different atomic masses (ie, they are isotopes of the same element), the substituent with the higher mass number has the higher priority. If two or more substituents are the same isotope of the same atom, continue to the next atom in each substituent chain and repeat until a difference is found. If no differences are found, then the central atom is not chiral.

After making assignments, position the lowest priority group at the back of the molecule (ie, a dashed bond) and determine whether the other groups are arranged in clockwise or counterclockwise order. Clockwise corresponds to an *R* configuration, and counterclockwise corresponds to an *S* configuration.

The passage states that ethylhexyl glycerin forms through an S_N2 reaction between glycerol and an ethylhexyl halide. The question shows ethylhexyl glycerin with a carbon bonded to deuterium where the halide (X) was prior to the reaction. This deuterated carbon is a chiral center that acted as the electrophile in an S_N2 reaction with glycerol, so the configuration of this carbon must have been inverted during the reaction. Based on the given rules, the deuterated ethylhexyl glycerin shown is in the *R* configuration. Therefore, the deuterated ethylhexyl halide used in the reaction must have been in the *S* configuration.

(Choices B and D) The product is in the *R* configuration, so the reactant must have been in the *S* configuration because electrophiles that are chiral centers undergo inversion of configuration in S_N2 reactions.

(Choice C) The ethylhexyl halide was in the *S* configuration, but this configuration was inverted, *not* preserved, by the S_N2 reaction.

Educational objective:

S_N2 reactions result in the inversion of configuration of the electrophile if the electrophile is a chiral center. An electrophile in the *S* configuration will adopt the *R* configuration after the reaction and vice versa.

Time spent:0 sec
Subject:
Organic Chemistry

QID:403690
System:
Introduction to Organic Chemistry

Two key ingredients found in many soaps are PEG 150 distearate and ethylhexyl glycerin. Both can be synthesized from glyceryl tristearate (Figure 1).

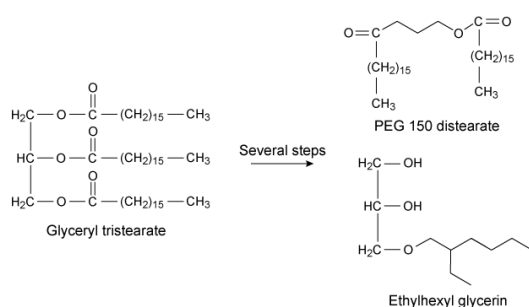


Figure 1 Formation of PEG 150 distearate and ethylhexyl glycerin from glyceryl tristearate

The first step in this process requires the saponification of glyceryl tristearate. The resulting stearate molecules can then react with PEG 150 in an acid-catalyzed Fischer esterification to form PEG 150 distearate. The glycerol produced during saponification is used as a nucleophile to form ethylhexyl glycerin through a base-catalyzed S_N2 reaction with an ethylhexyl halide.

Studies have shown that hand washing with soap can substantially reduce the number of bacteria present on hands and that washing hands before preparing food is particularly effective in reducing illness. Researchers examined the effect of washing for 20 seconds with different amounts of soap containing PEG 150 distearate and ethylhexyl glycerin and found an inverse correlation between the volume of soap used and the amount of bacteria recovered from the hands. The data are shown in Figure 2.

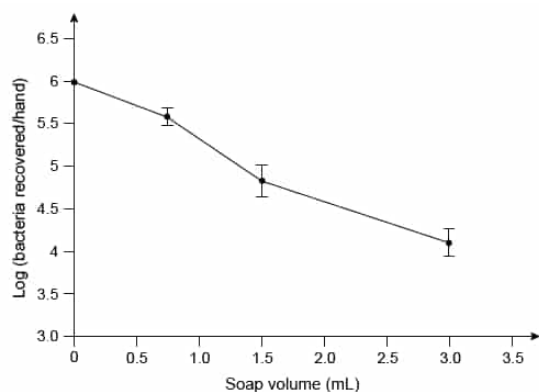


Figure 2 Bacteria recovered from hands after washing with varying amounts of soap

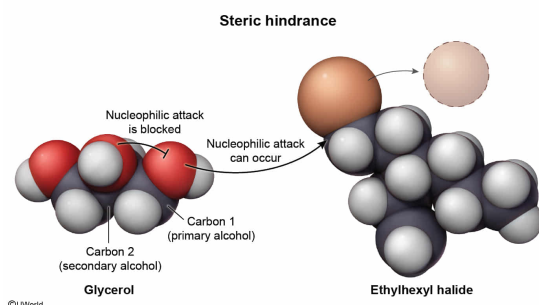
Why does the bond between glycerol and the 2-ethylhexyl group tend to form at glycerol's C1 hydroxyl group instead of the C2 hydroxyl group?

- ☐ A. Secondary alcohols are more likely to undergo S_N1 reactions than primary alcohols. [10%]
- ✓ ☒ B. Secondary alcohols are more sterically hindered than primary alcohols. [79%]
- ☐ C. Primary alcohols are less acidic than secondary alcohols. [1%]
- ☐ D. Primary alcohols are more electrophilic than secondary alcohols. [8%]

Correct

79% answered correctly

Explanation:



Steric hindrance can prevent reactions that would otherwise occur by physically blocking the access of one atom to another. Alcohols may be attached to primary, secondary, or tertiary carbon atoms, which are bonded to one, two, or three other carbons, respectively. These alcohols are referred to as **primary, secondary, and tertiary alcohols**. A **tertiary alcohol** is **more sterically hindered** than a **secondary alcohol** because it has more bonds to large atoms that can get in the way of interactions with other molecules. A secondary alcohol is **more hindered** than a **primary alcohol** for the same reason.

The passage states that ethylhexyl glycerin forms in an S_N2 reaction. The alcohol on carbon 2 of glycerol is a secondary alcohol, whereas carbon 1 is a primary alcohol, and the added **steric hindrance** at carbon 2 **prevents S_N2 reactions** from happening as readily as they do at carbon 1. Therefore, the C1 hydroxyl group reacts with an ethylhexyl halide more easily than the C2 hydroxyl group does.

(Choice A) Secondary alcohols are more likely than primary alcohols to undergo **S_N1 reactions**. However, the reaction of interest is an S_N2 reaction, for which primary alcohols are more efficient.

(Choice C) Primary alcohols are *more* acidic than secondary alcohols. This attribute contributes to their enhanced ability to react because deprotonation by a base makes the alcohol a better nucleophile.

(Choice D) The primary alcohol is acting as a **nucleophile**, so enhanced **electrophilicity** would make it a *worse* reactant.

Educational objective:

Sterically hindered chemical groups are physically inhibited from interacting with other molecules. For this reason, they are less capable of acting as nucleophiles in S_N2 reactions. Tertiary alcohols are more sterically hindered than secondary alcohols, which are more sterically hindered than primary alcohols.

Time spent: 0 sec

Subject:
Organic Chemistry

QID: 403691

System:
Introduction to Organic Chemistry

Two key ingredients found in many soaps are PEG 150 distearate and ethylhexyl glycerin. Both can be synthesized from glyceryl tristearate (Figure 1).

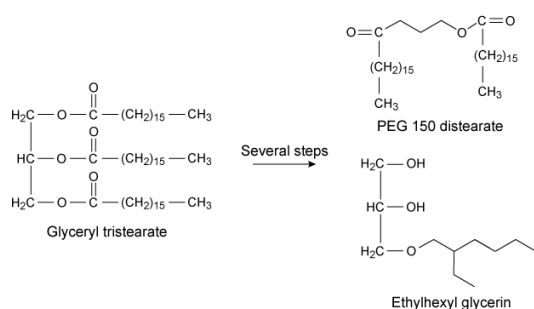


Figure 1 Formation of PEG 150 distearate and ethylhexyl glycerin from glyceryl tristearate

The first step in this process requires the saponification of glyceryl tristearate. The resulting stearate molecules can then react with PEG 150 in an acid-catalyzed Fischer esterification to form PEG 150 distearate. The glycerol produced during saponification is used as a nucleophile to form ethylhexyl glycerin through a base-catalyzed S_N2 reaction with an ethylhexyl halide.

Studies have shown that hand washing with soap can substantially reduce the number of bacteria present on hands and that washing hands before preparing food is particularly effective in reducing illness. Researchers examined the effect of washing for 20 seconds with different amounts of soap containing PEG 150 distearate and ethylhexyl glycerin and found an inverse correlation between the volume of soap used and the amount of bacteria recovered from the hands. The data are shown in Figure 2.

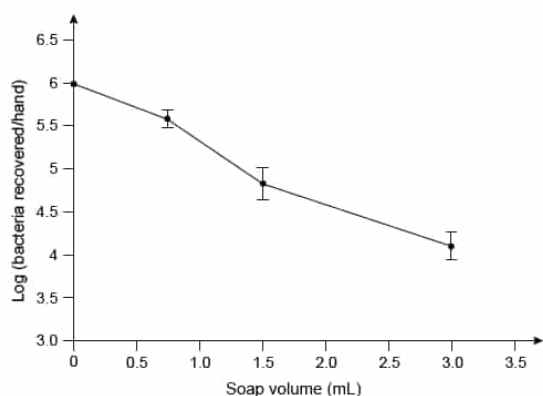


Figure 2 Bacteria recovered from hands after washing with varying amounts of soap

Acid helps catalyze the Fischer esterification reaction by doing all of the following EXCEPT:

- ☒ A. enhancing the nucleophilicity of alcohols. [44%]
- ☐ B. forming a resonance-stabilized intermediate. [24%]
- ☐ C. creating a good leaving group. [12%]
- ☐ D. increasing carbonyl electrophilicity. [18%]

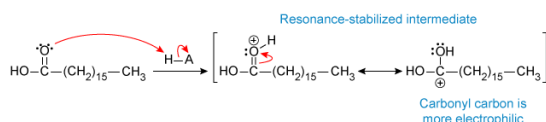
Correct

43% answered correctly

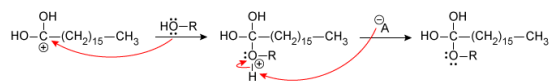
Explanation:

A Fischer esterification is the acid-catalyzed condensation of a **carboxylic acid** and an **alcohol** into an **ester**. **Strong acids** can **donate** their **protons** to other molecules to **catalyze** a reaction. In the Fischer esterification, acidic protonation enhances the rate of esterification.

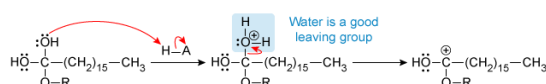
In the **mechanism** for the Fischer esterification, **protonation** of the carbonyl oxygen leads to the formation of a **resonance-stabilized intermediate (Choice B)**. A resonance form of this intermediate has a positive formal charge on the carbonyl carbon, which **enhances its electrophilicity (Choice D)**.



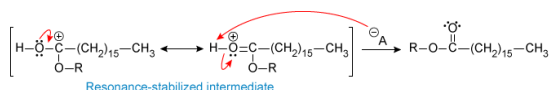
The alcohol nucleophile attacks the electrophilic carbon. Subsequent deprotonation regenerates the acid catalyst.



The protonation of a hydroxyl converts it into a **good leaving group (Choice C)**, which is eliminated as a molecule of **water**.



The resulting cation is a **resonance-stabilized intermediate**. Deprotonation of the carbonyl oxygen again regenerates the acid catalyst and forms the ester product.



Acids do not generally increase the nucleophilicity of a compound, as a nucleophile lone pair is attracted to acidic protons. Consequently, acids tend to protonate nucleophiles and make them *less* nucleophilic.

Educational objective:

Acids catalyze many reactions by donating protons to a reactant. Protonation enhances electrophiles and increases the stability of leaving groups. Acids generally decrease the nucleophilicity of molecules.

Time spent: 0 sec

Subject:

Organic Chemistry

QID: 403692

System:

Introduction to Organic Chemistry

Two key ingredients found in many soaps are PEG 150 distearate and ethylhexyl glycerin. Both can be synthesized from glyceryl tristearate (Figure 1).

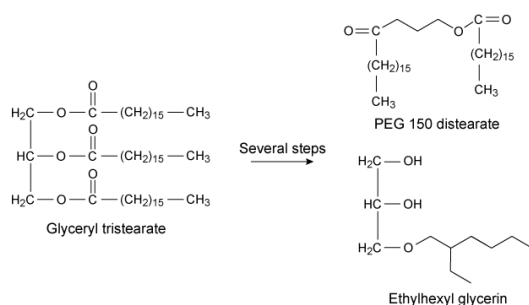


Figure 1 Formation of PEG 150 distearate and ethylhexyl glycerin from glyceryl tristearate

The first step in this process requires the saponification of glyceryl tristearate. The resulting stearate molecules can then react with PEG 150 in an acid-catalyzed Fischer esterification to form PEG 150 distearate. The glycerol produced during saponification is used as a nucleophile to form ethylhexyl glycerin through a base-catalyzed S_N2 reaction with an ethylhexyl halide.

Studies have shown that hand washing with soap can substantially reduce the number of bacteria present on hands and that washing hands before preparing food is particularly effective in reducing illness. Researchers examined the effect of washing for 20 seconds with different amounts of soap containing PEG 150 distearate and ethylhexyl glycerin and found an inverse correlation between the volume of soap used and the amount of bacteria recovered from the hands. The data are shown in Figure 2.

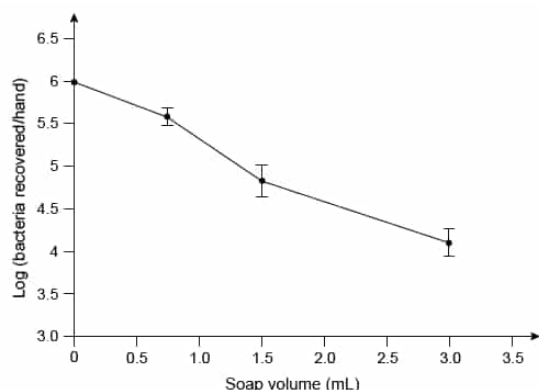


Figure 2 Bacteria recovered from hands after washing with varying amounts of soap

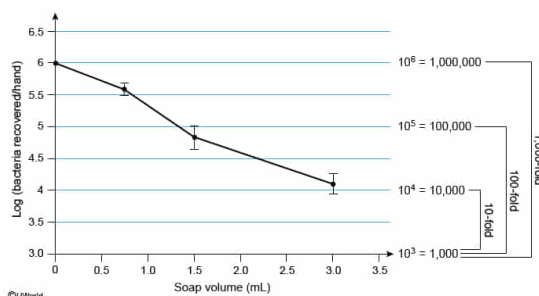
Compared to hands washed with no soap, washing hands with 3.0 mL of soap reduced bacteria recovered per hand by a factor of approximately:

- ☐ A. $\frac{1}{3}$ [20%]
☐ B. 2 [8%]
☐ C. 10 [4%]
☒ D. 100 [66%]

Correct

66% answered correctly

Explanation:



Graphical data are frequently displayed on one of two scales: **linear or logarithmic**. Linear scales are useful for displaying relatively small changes whereas **logarithmic (log) scales** are useful when the **changes** are large, spanning **multiple orders of magnitude**. Unless a different base is specified, each step up on a log scale represents a 10-fold increase in the measured parameter, and each step down is a 10-fold decrease. Therefore, **one step** is a **10-fold change**, **two steps** is a **100-fold change**, three steps is a 1,000-fold change, and so on.

Figure 2 is a semi-log plot, with the y-axis on a log scale and the x-axis on a linear scale. It shows that when hands are washed without soap, approximately 10^6 (or 1,000,000) bacteria were recovered on average from volunteers' hands because $\log(10^6) = 6$. However, after washing with 3 mL of soap, approximately 10^4 (or approximately 10,000) bacteria were recovered on average, giving approximately a 100-fold reduction in the number of bacteria recovered. Note that although the error bars indicate that different fold reductions were observed in different trials, and the average reduction was slightly less than 100-fold, the question asks only for an *approximate* level.

(Choice A) The values represented on the y-axis of the graph are approximately 6 without soap and 4 with 3.0 mL of soap. A change from 6 to 4 represents a 1/3 reduction on a linear scale, but the graph uses a log scale, in which this change represents approximately a 100-fold reduction.

(Choice B) The *difference* between 6 and 4 is 2, but this does not represent the factor by which recovered bacteria changed after hand washing.

(Choice C) On a log scale, a 10-fold reduction would appear as a difference of one step. An approximate 10-fold reduction was observed when hands were washed with 1.5 mL of soap instead of 3.0 mL of soap.

Educational objective:

Graphs may display data on a linear scale, a logarithmic (log) scale, or a combination with one scale on each axis, depending on the size of the data being displayed. Log scales display changes in order of magnitude: one step corresponds to a 10-fold change, two steps a 100-fold change, and so on.

Time spent: 0 sec

Subject:
Organic Chemistry

QID: 403693

System:
Introduction to Organic Chemistry

微信Mai_ange